

4 Demand

4.1 Individual Demand

We will now show how to use the theory of preferences from last chapter, to derive an individual's demand curve. Remember that the consumer's budget line can change because of changes in prices or because of changes in income. Here, we will assume that the preferences themselves do not change. This makes us able to derive both the demand curve that we used in Chapter 2, and the so-called Engel curve, which shows how demand depends on income.

4.1.1 The Individual Demand Curve

As we showed in Chapter 3, it is possible to find the point of utility maximization if one knows a consumer's preferences, the prices of the goods, and her budget. Let us now do that, but vary the price of good 1 and see what effect that has on, q_1 , the quantity demanded.

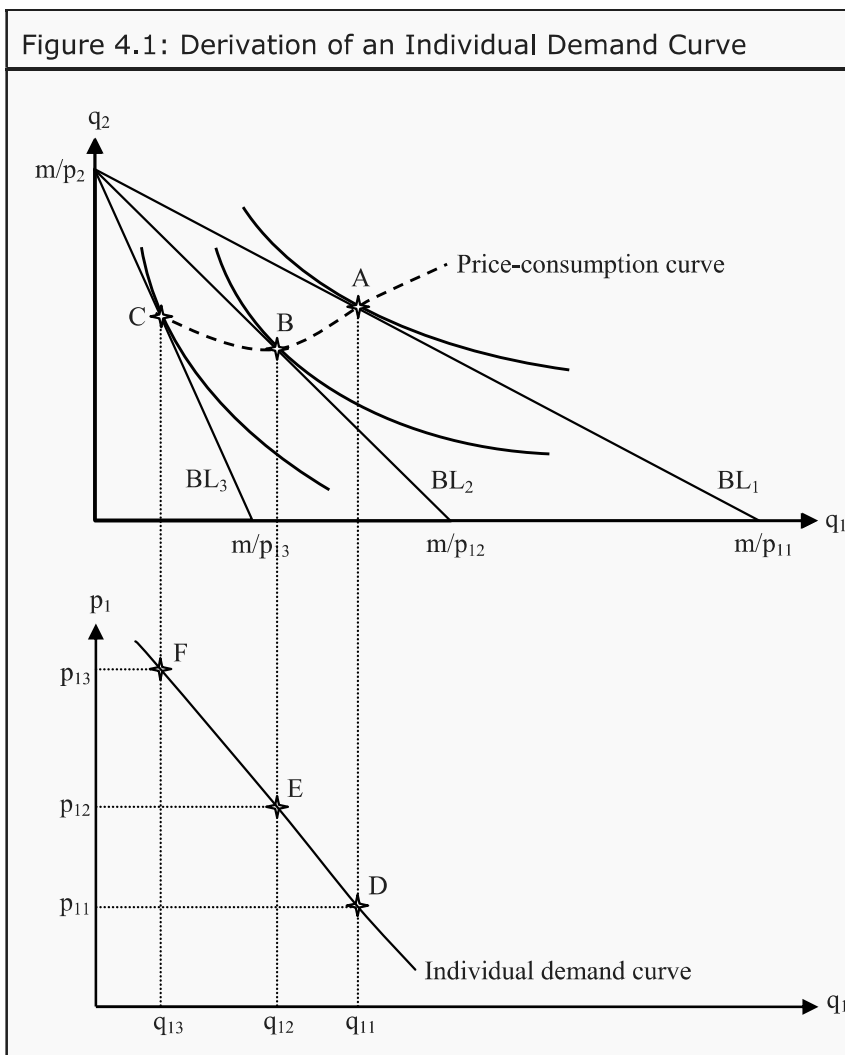
Suppose we hold the price of good 2 (which you can think of as "all other goods") constant. Then the effect of varying the price of good 1 will be that the budget line rotates about the intercept on the Y-axis and intersects the X-axis at different points m/p_{1i} , where p_{1i} is the price one has chosen for good 1.

Look at the upper part of Figure 4.1. Suppose the price of good 1 is initially p_{11} . Then the budget line is BL_1 . We find the indifference curve that just touches that budget line and label the point where it does so, point A. If we would raise the price of good 1 to p_{12} , the possible choices become limited to BL_2 (that intersects the X-axis in m/p_{12}) and then the consumer maximizes her utility in point B. If we continue to raise the price to p_{13} , and repeat the maximization, we get point C. If we would repeat this procedure for all possible prices, we would get a curve that is called the **price-consumption curve**. It shows how the optimal choice of quantity of good 1 varies with the price of that good, given that preferences, other prices and the income are held constant.

Price-consumption curve:
A curve that shows how a consumer chooses to consume at different prices.

As you can see in the figure, the consumer will usually buy less of the good when the price increases. This is, however, not necessary. To see that, imagine that the indifference curve that runs through point B had been steeper. If it had been steep enough, it would touch BL_2 so far to the right that it would also be to the right of point A.

Now we want to find the demand curve for good 1. To that end, we indicate the prices we used for good 1 on the Y-axis in the lower graph of the figure, i.e. p_{11} , p_{12} , and p_{13} . Then we check which are the corresponding quantities demanded in the upper graph, at points A, B, and C, and indicate them on the X-axis in the lower diagram. (Note that both diagrams have q_1 on the X-axis.) After that, we find the points where the quantities and the corresponding prices in the lower diagram intersect, the points labeled D, E, and F. Finally, we draw a line through those points and fill in for all those numerous points for which we have not done the analysis. This curve is the individual's demand curve for good 1.



4.1.2 The Engel Curve

In the previous section, we showed how it is possible to derive the relation between the price and the quantity demanded for a certain good. Now, we will instead show how to derive the relation between *income* and the quantity demanded. The resulting curve is called the **Engel curve**.

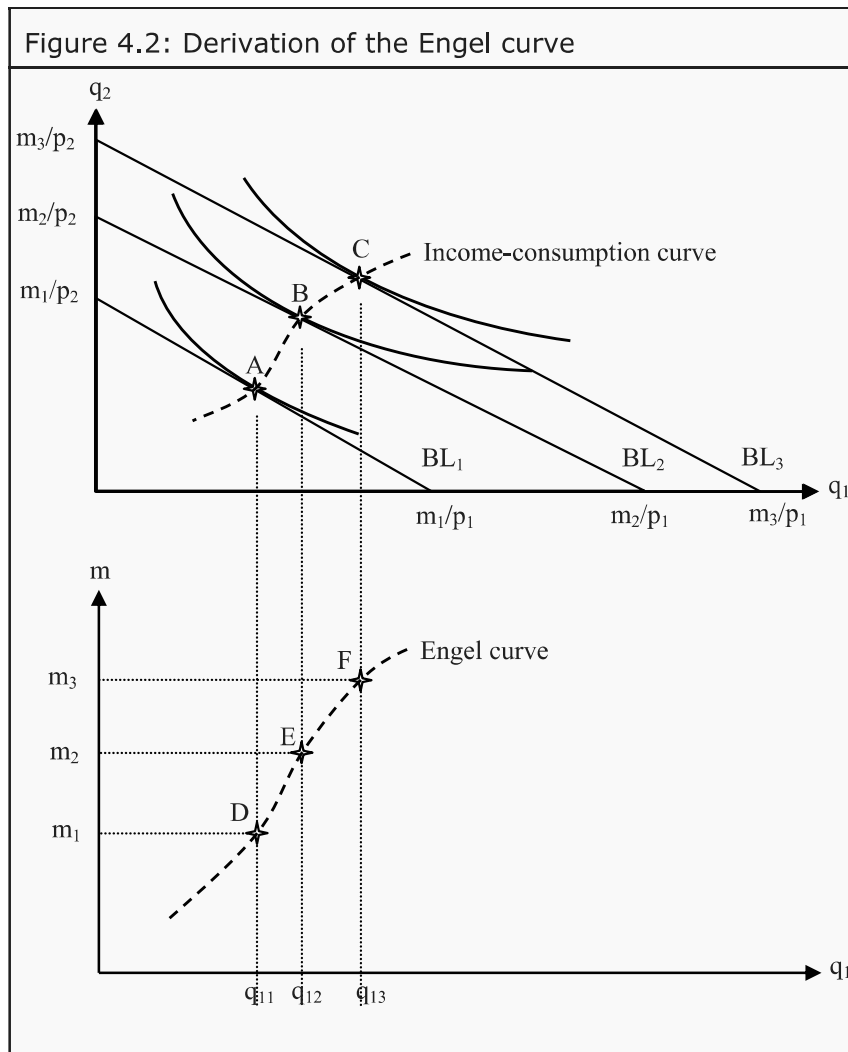
Look at Figure 4.2. Just as in the previous case, we start with the individual's maximization problem where she must choose quantities of good 1 and good 2. (Again, think of good 2 as "all other goods.") However, instead of varying the price, we now vary the income m . This means that the budget line will shift outwards for higher incomes and inwards for lower incomes. We assume that preferences and prices are unchanged. For the increasingly higher incomes m_1 , m_2 , and m_3 , the budget lines become BL_1 , BL_2 , and BL_3 .

In the same way as before, we find the utility maximization points for each budget line: points A, B, and C. If we would do that for all possible incomes, we would get the so-called **income-consumption curve**. That curve shows the optimal consumption of good 1 and good 2 at different incomes, given preferences and prices.

Engel curve: A curve that shows the relation between income and quantity demanded. Compare to the demand curve.

Income-consumption curve: A curve that shows the relation between income and consumption.

Similarly to before, we indicate the quantities that correspond to points A, B, and C, i.e. q_{11} , q_{12} , and q_{13} in the diagram below. Then we indicate the incomes m_1 , m_2 , and m_3 on the Y-axis, and the points where the incomes intersect the corresponding quantities: points D, E, and F. Thereafter, we draw a line through the points of intersection, as it would probably have looked if we had performed the same procedure for the points in between. The resulting curve is the so-called Engel curve, and it shows how the optimal consumption of good 1 varies with the income, given preferences and prices.



4.2 Market Demand

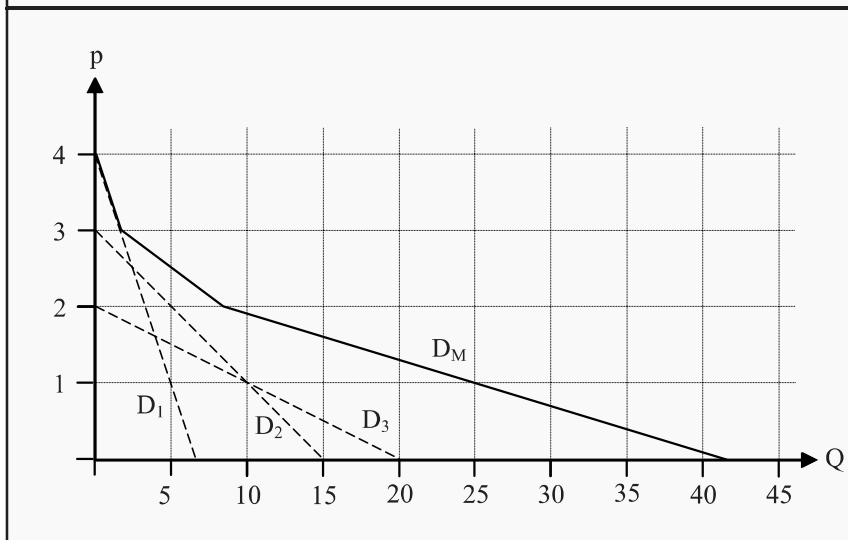
The market's demand consists of all individuals' demand. To find the market demand curve, we have to sum up the demand of all individuals for each price.

Suppose, for instance, that we have found demand curves for three different individuals, and that these three individuals together are the whole market. In Figure 4.3, their demand curves (for simplicity, they are all straight lines) are labeled D_1 , D_2 , and D_3 .

If the price of the good is 4, all individuals demand a quantity of 0, but at a price of 3, the first individual demands 2 units. Since the others do not demand anything, the market's total demand is those 2 units. For prices between 3 and 4, the market's demand coincides with D_1 , i.e. the demand curve of the first individual. (A straight line from point (0,4) to point (2,3))

When the price is 2, the first individual demands 4 units and the second demands 5 units. The market's total demand is then 9 units. For prices between 2 and 3, total demand is $D_1 + D_2$. It will then be a straight line beginning in point (2,3) and ending in point (9,2).

Figure 4.3: Market Demand



Question:

What do Skype and Spotify have in common with color screen graphics and the computer mouse?

– They are all Swedish inventions.

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When the price is close to 0, all individuals demand it: The first demands 7 units, the second demands 15 and the third demands 20. Total demand is then 42 units. For prices between 0 and 2, total market demand is $D_1 + D_2 + D_3$. It will then be a straight line starting in point (9,2) and ending in point (42,0). Note that we should not really allow a price of 0. Demand would then be infinite as more of the good is, by assumption, always better.

The **market demand curve**, D_M , will consequently be the sum of the individual demand curves. If the individual demand curves are straight lines, the market demand curve will become a succession of straight lines, where a break signals that a new consumer starts demanding the good at that price.

Market demand curve: A curve that shows the demand for the whole market at different prices.

4.3 Elasticity

Suppose we want to study the effects a price change has on the demand of a good. It is practical to do that in terms of percentages: If the price rises by one percent, how many percentages will demand change?

More generally, one can study how many percent any one variable changes when another variable changes by one percent. This is called **elasticity**. The types of elasticity that are used the most are price elasticity, income elasticity and cross-price elasticity.

Elasticity: A measure of how sensitive a variable is to changes in another variable.

4.3.1 Price Elasticity

Price elasticity (of demand) is how many percent demand changes if the price changes one percent. We use the notation e_p for price elasticity, Q for quantity demanded, ΔQ for the change in quantity demanded, p for the price, and Δp for the change in price. The price elasticity can then be calculated as

$$e_p = \frac{\Delta Q / Q}{\Delta p / p}$$

Note that the expression in the numerator is the relative change in quantity (relative to the level) and the expression in the denominator is the relative change in price.

The elasticity is usually different depending on where on the demand curve it is calculated, even if the demand curve is a straight line. To see that, look at Figure 4.4. We start with point A where the price is 15 and the quantity demanded is 5. For simplicity, we choose Δp to be 1. If the price increases with 1, the quantity demanded decreases by 1 ($\Delta Q = -1$), i.e. we move one step upwards and one step to the left following the arrows. The price elasticity at point A is consequently $e_p = (-1/5)/(1/15) = -3$. If we perform the same exercise at point B, we get that $e_p = (-1/16)/(1/4) = -0.25$.